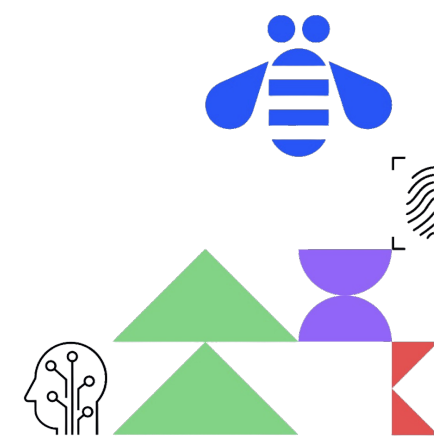


Achieving Crypto Agility with IBM Guardium Cryptography Manager

Quantum Safe & Cryptography
Management Hands-on
An IBM TechXchange Workshop

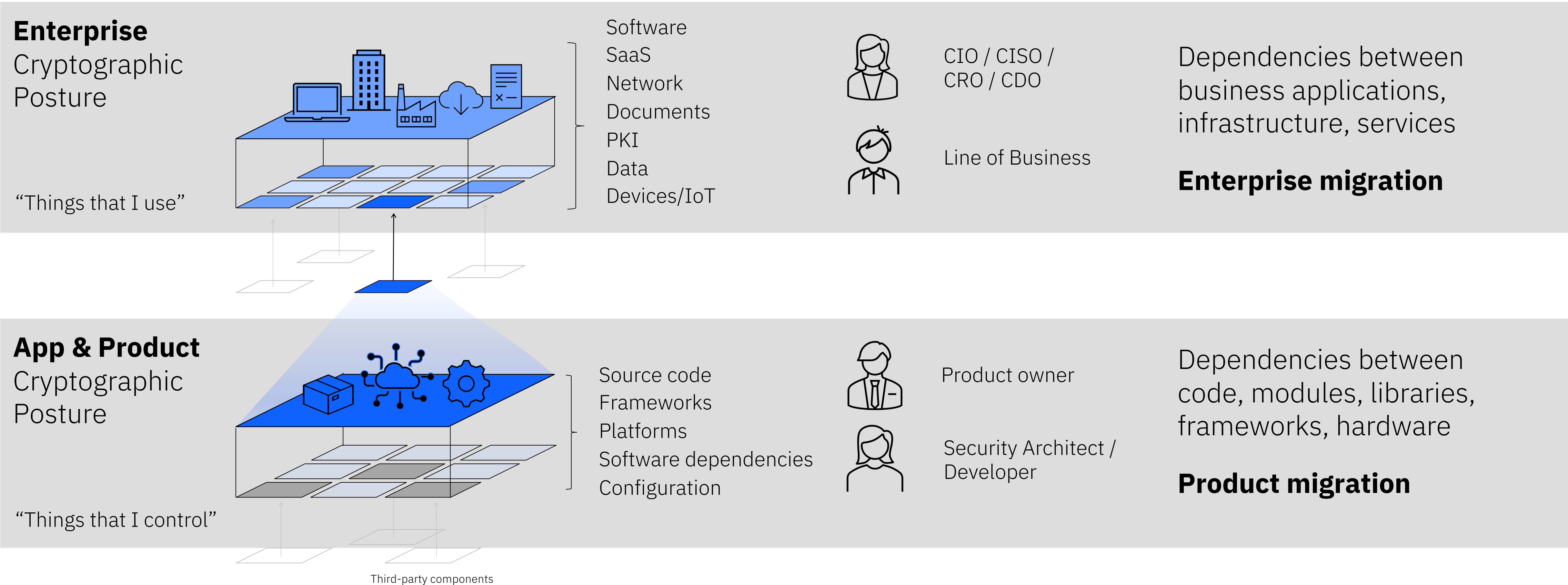


2. September 2026

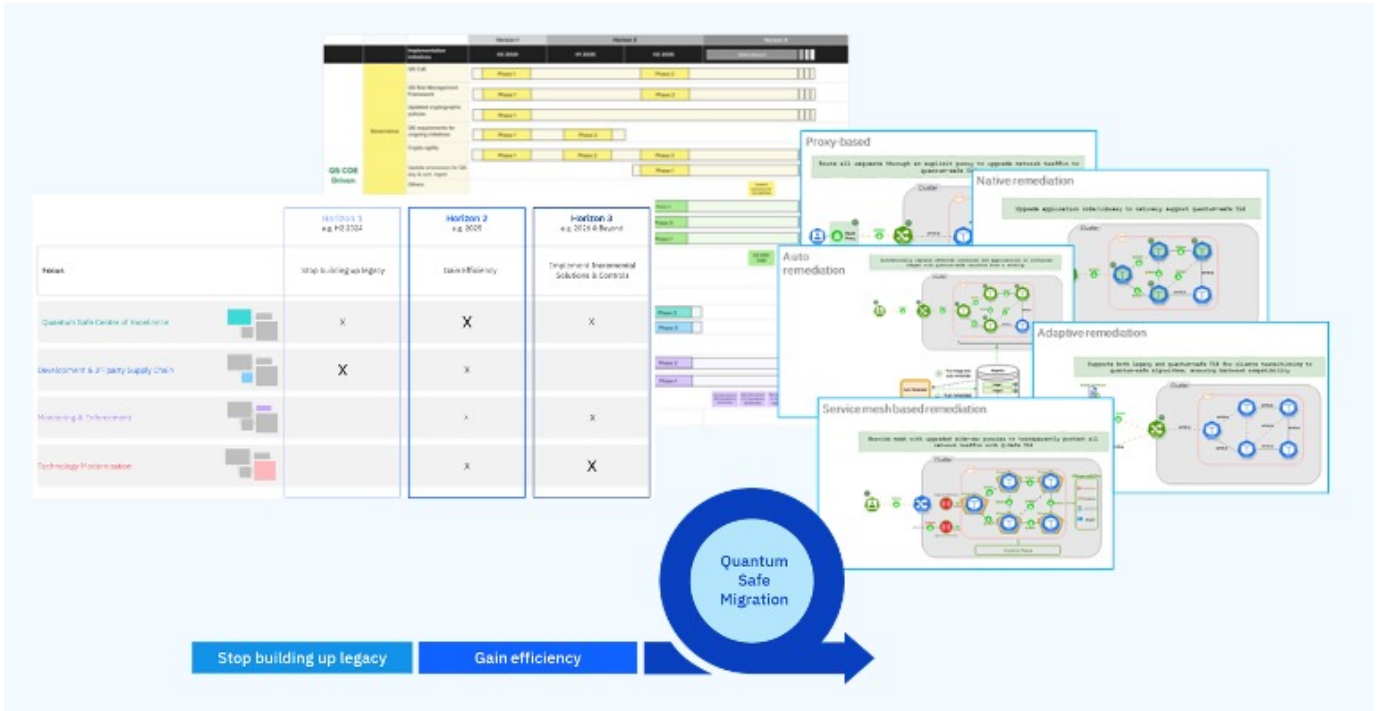
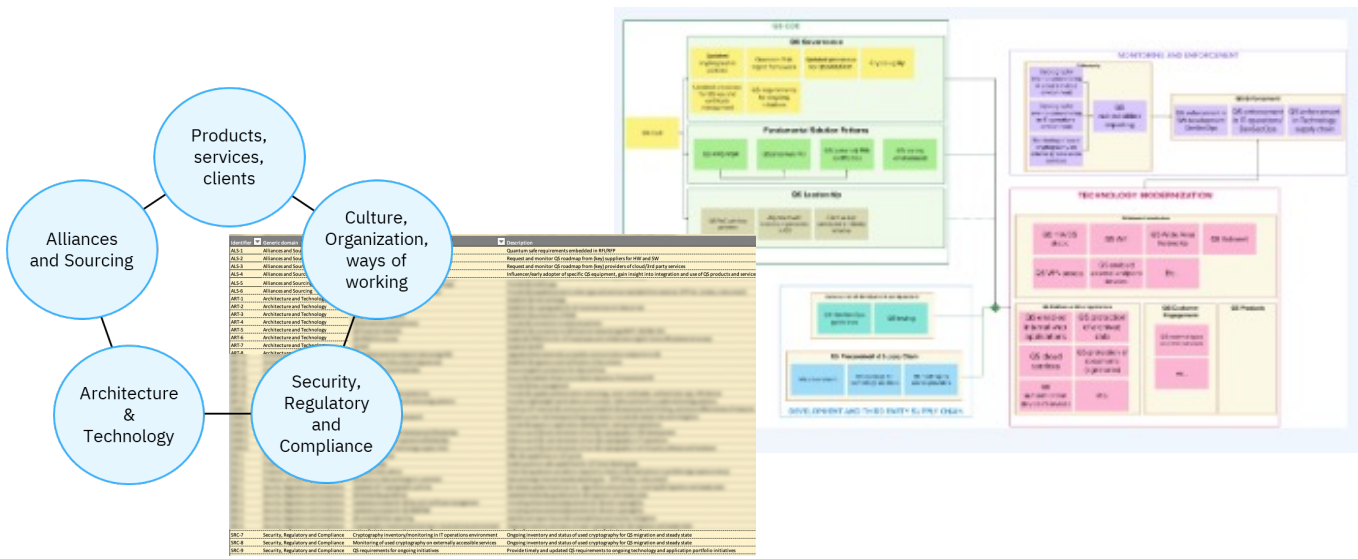
Vladimir Atanaskovik
IBM Data & AI Security
[E-mail: vladimir@de.ibm.com](mailto:vladimir@de.ibm.com)



How to manage cryptography transition? It's all about context!



Path to Quantum Safe transformation



Understand the situation

Define the target

Execute transformation

- Quantum Safe deep dive and awareness
- Map out affected assets, and their dependencies (“QS heatmap”)
- Identification of potential urgent actions
- Outline of quantum safe next steps

INVENTORY OF CRYPTOGRAPHY RELATED BUSINESS AND TECHNOLOGY ASSETS

- Identify quantum safe related capabilities (target state)
- Prioritize actions and initiate urgent measures
- Define implementation approach and governance
- Set up Quantum Safe initiative

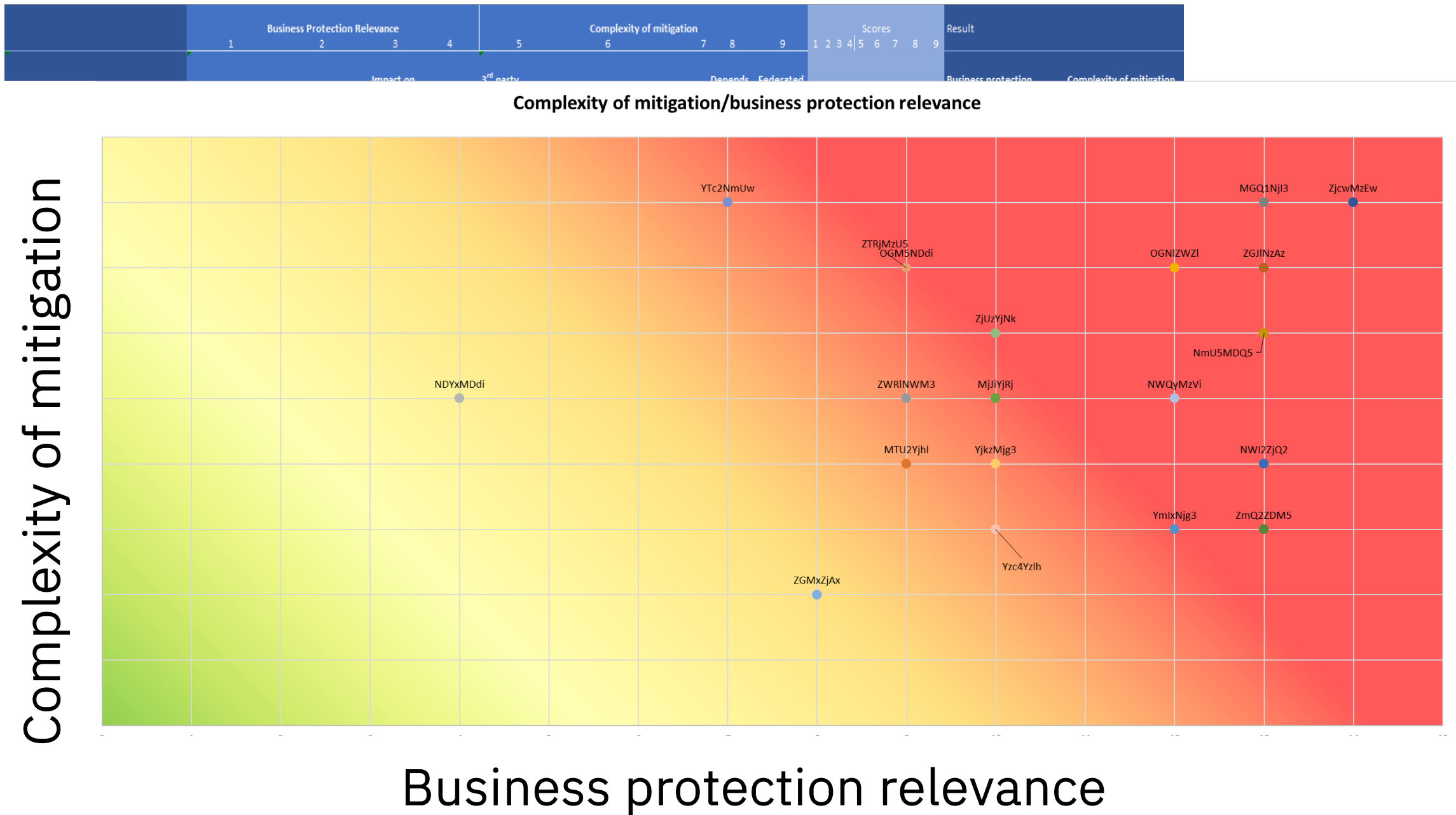
TARGET AND PLAN

- Implement actions in line with ways of working, internal initiatives and ecosystem
- Apply predefined and validated patterns and artefacts wherever possible
- Monitor technology progress and adjust approach where needed

EFFICIENT & AGILE EXECUTION

Path to Quantum Safe transformation – Heatmap, Fields of Actions

Quantum Safe Heatmap

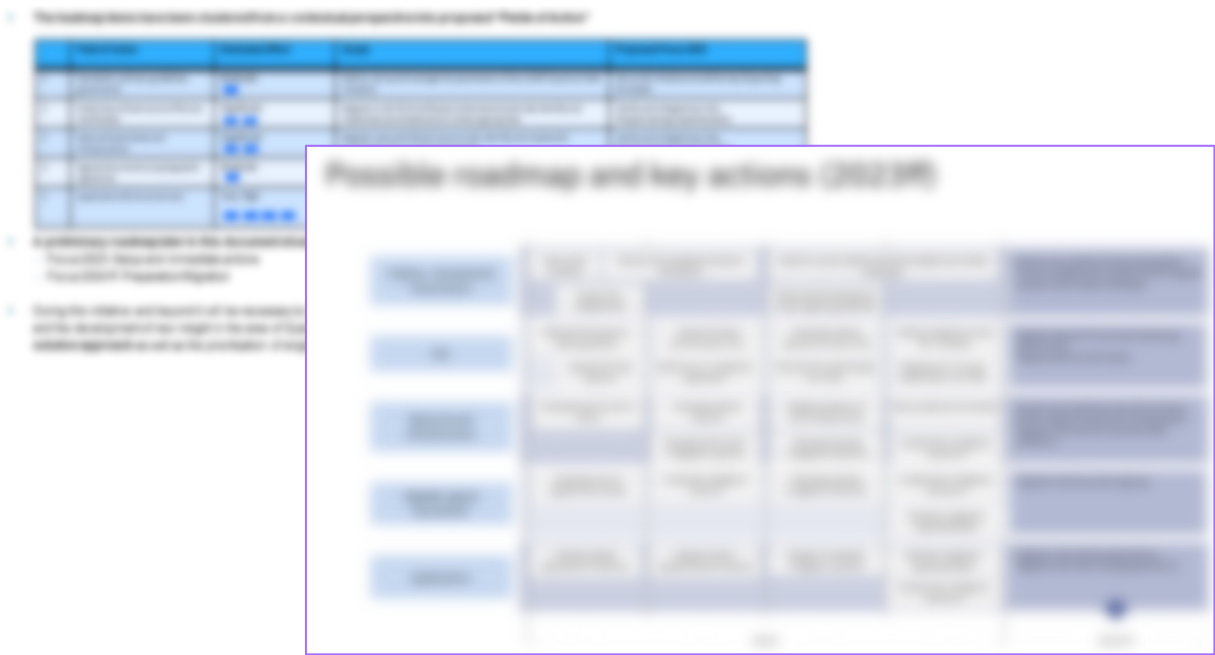


Elements cover applications, tools, network, infrastructure, (cryptographic) governance, business & 3rd party dependencies, etc.

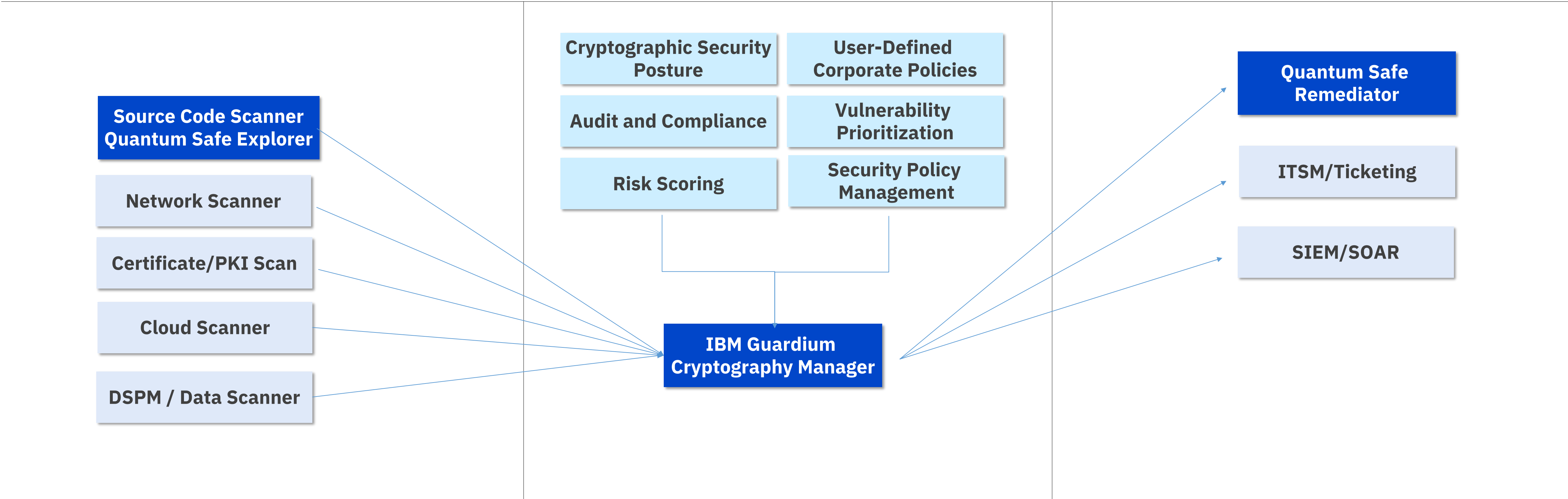
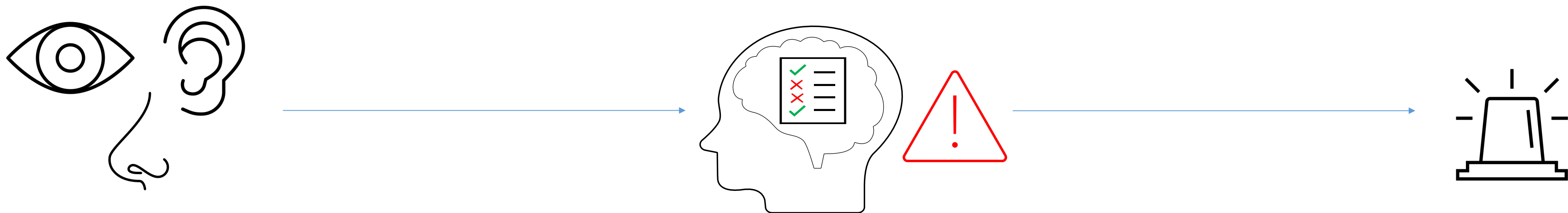
INVENTORY OF CRYPTOGRAPHY RELATED BUSINESS AND TECHNOLOGY ASSETS

Quantum Safe Fields of Action

- ✓ Fields of Action focus on urgent areas (in terms of business risks) and non-regret actions for Quantum Safe mitigation.
- ✓ Defines specific areas for remediation, estimated effort and concrete list of tasks to start with.

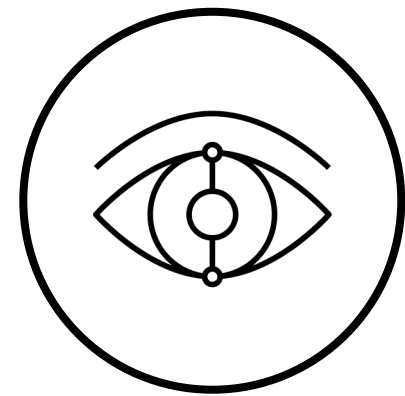


IBM quantum safe technologies (Crypto Agility)



IBM Guardium Cryptography Manager

Protecting today. Preparing for tomorrow.

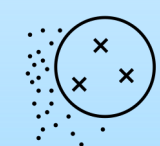


Discover and Inventory

Automatically and continuously discover and inventory your cryptographic landscape



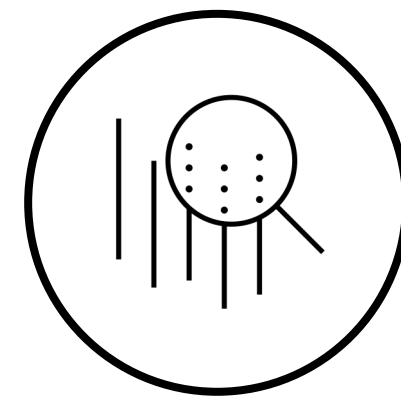
Identify cryptographic objects and IT assets



Detect unknown or shadow cryptography



Map dependencies, ownership, and usage

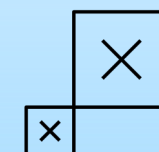


Assess and Comply

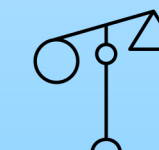
Assess your cryptographic posture to prioritize vulnerabilities and maintain compliance



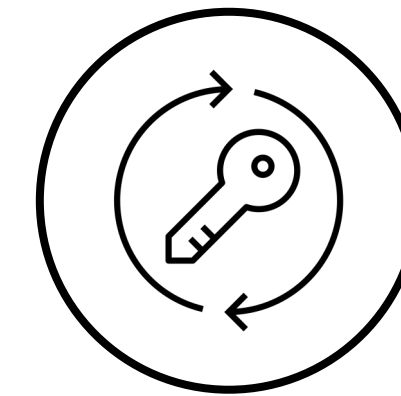
Assess post-quantum cryptography risk



Flag outdated or vulnerable algorithms

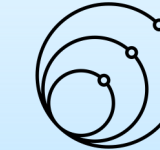


Generate audit-ready reports

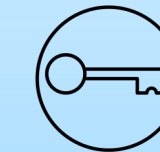


Manage Lifecycle

Simplify and automate cryptographic object management



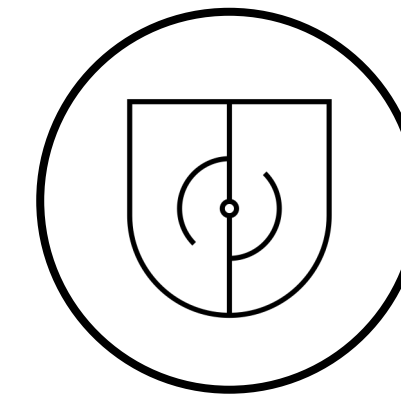
Centralize lifecycle operations in one tool



Automate key generation and certificate renewal



Enforce consistent policies across the organization

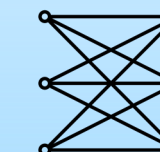


Protect and Adapt

Encrypt data and adopt a crypto-agile architecture



Utilize encryption to protect sensitive data

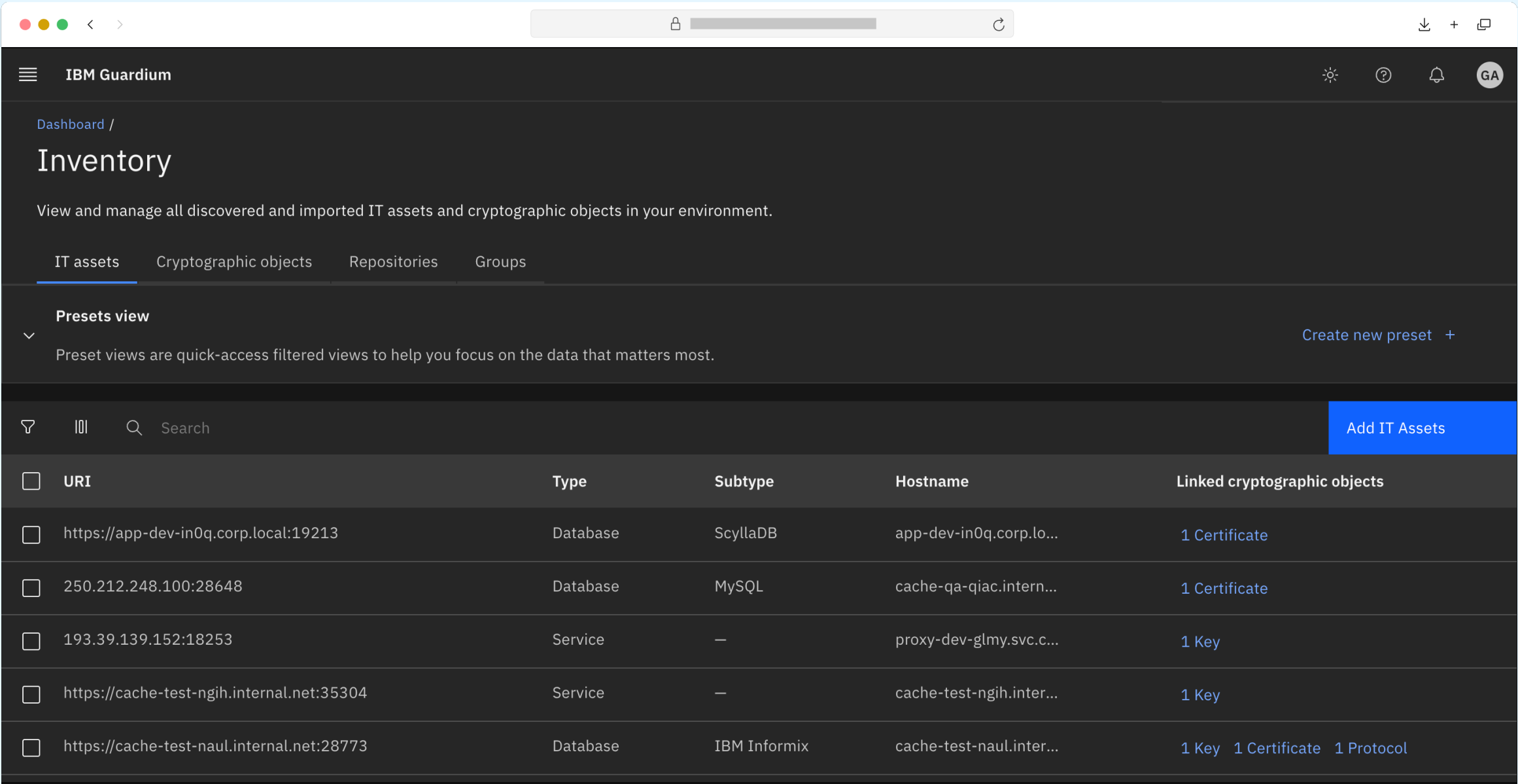
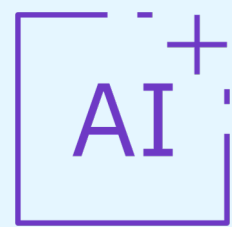


Assess the efficiency and feasibility of PQC algorithms



Protect against "harvest now, decrypt later" attacks

Gain comprehensive visibility through discovery and inventory



1

Automatically discover and catalog encryption keys, certificates, algorithms, and datastores across all environments

2

Gain clear visibility with a comprehensive, consolidated dashboard of your organization’s inventory of cryptographic objects

3

Automate enforcement of cryptographic policies and easily generate audit-ready reports to demonstrate compliance

4

Detect weak, expired, or misconfigured assets through continuous threat monitoring and cryptographic posture assessment

How to tackle both today's challenges and build resilience against the quantum threat:

Crypto-Agility

[krip-toh-*uh*-jil-i-tee]

noun

“The ability to rapidly adapt cryptographic mechanisms in response to threats, technological advances, or vulnerabilities – without disrupting infrastructure or business processes”

Our mission

Bring useful quantum
computing to the world

Make the world
quantum safe

Learn more about IBM Quantum Safe solutions: [Explore IBM Quantum Safe](#)

